Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 1 Quiz 1 - Kinematics in One Dimension

Mock Exam D · Form 2: Multi-step reasoning

Multiple Choice Questions: Select the best answer. (1 point each)

1. A motion sensor records a cart starting at +3 m/s. Acceleration is +5 m/s² for 2 s, then -2 m/s² for another 2 s. What is its total displacement?

(A) 38 m

(B) -38 m

(C) 76 m

(D) 19 m

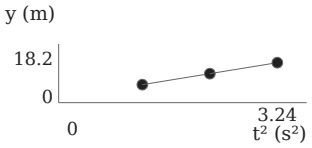
2. A sensor records downward position for a weight dropped from rest. Use g=10 m/s². Release from the same coordinate where gravity is halved. What is the new ratio of vertical change to the same horizontal change?

(A) 3.55

(B) 0.4

(C) 5.65

(D) 2.5



3. A cart starts at +7 m/s and maintains a = -1 m/s². Over T = v0/|a| + 2 s (approximately 9 s), what is the time when direction reverses?

(A) 7 s

(B) -7 s

(C) 14 s

(D) 3.5 s

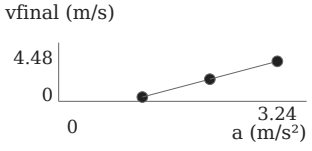
4. A cart initially moving at -2 m/s receives a constant acceleration lasting 2 s each trial. a (m/s²) -> vfinal (m/s): (1.24, 0.48); (2.24, 2.48); (3.24, 4.48). Which comparison distinguishes vertical zero bias from slope error?

(A) Compare observed minus predicted residuals at separated settings.

(B) Remove units before comparing.

(C) Force every fit through the origin.

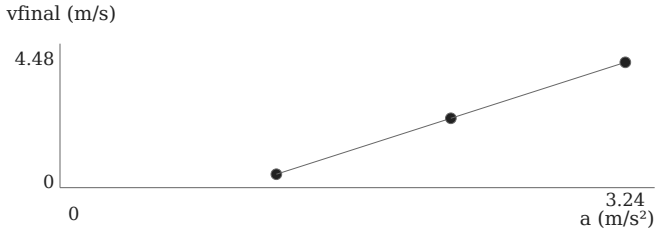
(D) Repeat at only one setting.



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A cart initially moving at -2 m/s receives a constant acceleration lasting 2 s each trial. a (m/s²) -> vfinal (m/s): (1.24, 0.48); (2.24, 2.48); (3.24, 4.48).

a. For the velocity-area experiment setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Double pulse duration without changing initial velocity. Predict the new slope and intercept and explain physically what changed. **(3 pt)**

6. Photogates record the position of a cart traveling at constant velocity. t (s) -> x (m): (0.62, 6.76); (1.12, 5.76); (1.62, 4.76).

a. Build the model for track photogates from physics and predict both slope and intercept. **(2 pt)**

b. Reverse the positive position direction only. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. A sensor records downward position for a weight dropped from rest. Use g=10 m/s². t² (s²) -> y (m): (1.24, 8.2); (2.24, 13.2); (3.24, 18.2). Build the model for drop sensor from physics and predict both slope and intercept. **(3 pt)**